

# Numbers and Sets

**ECON 441: Introduction to Mathematical Economics**  
**Lecture 1**

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# Introductions

- Your preferred name and pronouns
- Why are you here?
  - Undergrads: why you signed up for this class
  - Grad students: why you're doing the MA
- The last math class you took, and roughly when
- Your comfort food

# Real-Number System

- **Integers:**

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

- **Fractions:**

$\frac{1}{2}, \frac{3}{5}, -\frac{2}{3}$

- **Rational numbers:** ratio of integers
- **Are fractions rational numbers? What about integers?**

# Real-Number System

- **Rational numbers:** ratio of integers
  - “terminating or repeating decimal”
  - Example.  $\frac{1}{3} = 0.333$ ,  $\frac{1}{4} = 0.25$
- **Irrational numbers:** cannot be expressed as a ratio of two integers
  - “nonterminating or nonrepeating decimal”
  - Example.  $\sqrt{2} = 1.4142$ ,  $\pi = 3.1415$
- **Real numbers ( $\mathbb{R}$ ):** rational and irrational

# Sets

- A **set** is a collection of distinct objects.

$$A = \{brownies, icecream, pizza, ramen\}$$

- $pizza \in A$ ,  $\in$  stands for 'is in'
- Other ways of writing a set

$$B = \{x | x \text{ is a positive integer}\}$$

$$C = \{x | 1 < x < 5\}$$

# Set Relations

- Equivalence (=)

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{pizza, ramen, icecream, brownies\}$$

$$A = B$$

- Subset ( $\subset$ )

$$C = \{pizza, ramen\}$$

$C \neq A$  but  $C \subset A$ . Note:  $A \supset C$  is equivalent. **Is  $A \subset B$ ?**

# Set Relations

- Disjoint sets

$$A = \{brownies, icecream, pizza, ramen\}$$

$$D = \{salad, fruits\}$$

- Neither but still related

$$A = \{brownies, icecream, pizza, ramen\}$$

$$E = \{salad, fruits, icecream\}$$

# Set Relations

- $\emptyset$ : empty or null set
- What are all possible subsets of  $S = \{a, b, c\}$ ?
  - $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$
  - Always  $2^n$  subsets. Here  $n = 3$ , so 8 subsets.



# Set Operations

- **Union:**  $A \cup B$ , elements in either  $A$  **or**  $B$
- **Intersection:**  $A \cap B$ , elements in both  $A$  **and**  $B$
- Example:

$A = \{brownies, icecream, pizza, ramen\}$

$B = \{salad, fruits, icecream\}$

-  $A \cup B =$

-  $A \cap B =$

# Set Operations

- **Union:**  $A \cup B$ , elements in either  $A$  or  $B$
- **Intersection:**  $A \cap B$ , elements in both  $A$  and  $B$
- What about

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{salad, fruits\}$$

$$- A \cup B =$$

$$- A \cap B =$$

# Set Operations

- **Complement of  $A$ :**  $\tilde{A}$  or  $A^c$ , 'not  $A$ '
- Universal set  $U$  (context specific) then:

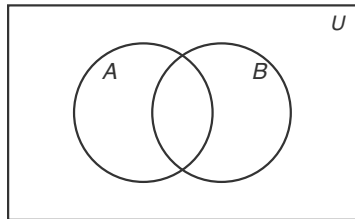
$$A^c = \{x | x \in U \text{ and } x \notin A\}$$

Example.  $U = \{1, e, f, 2\}$ ,  $A = \{1, 2\}$ , then  $A^c = \{e, f\}$ .

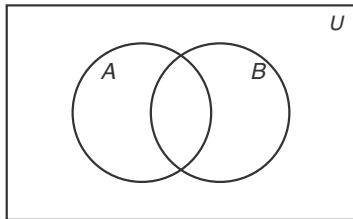
# Set Operations: Venn Diagrams

Shade the following regions:

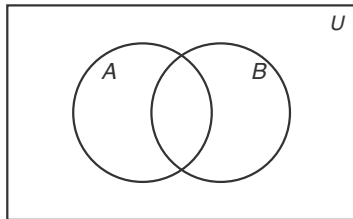
$$A \cup B$$



$$A \cap B$$



$$A^c$$



# Laws of Set Operations

- Commutative law

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

- Distributive law

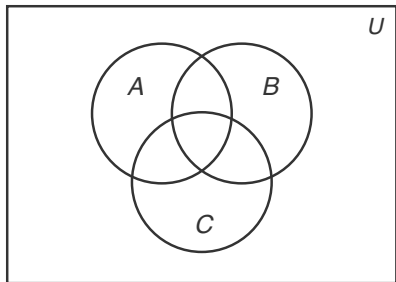
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

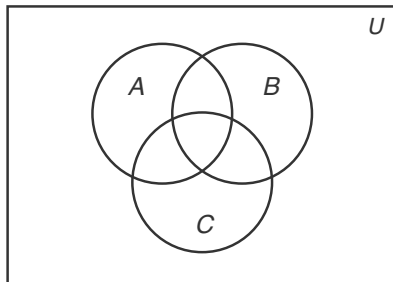
## Distributive law

Verify by shading:  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$A$



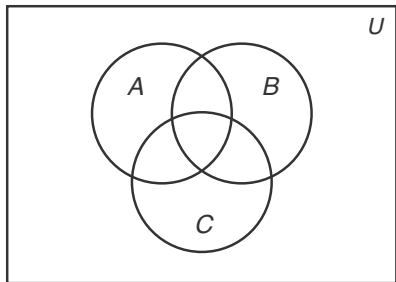
$B \cap C$



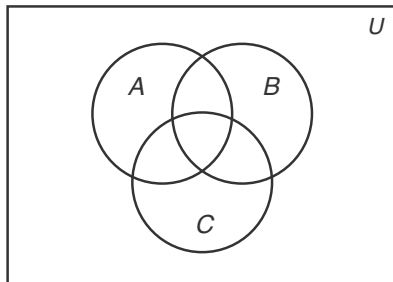
# Distributive law

Now the right-hand side,  $(A \cup B) \cap (A \cup C)$ :

$A \cup B$



$A \cup C$



# Ordered Sets

- We said order does not matter for sets
- But we can have ordered sets where

$$(a, b) \neq (b, a) \text{ unless } a = b$$

- Ordered pairs, triples, ...
  - Example.  $(age, weight)$ ,  $(22, 120)$  different from  $(120, 22)$



# Cartesian Product

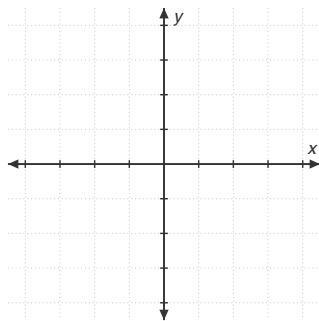
$$A = \{1, 2\} \quad B = \{3, 4\}$$

Cartesian Product: set of all possible ordered pairs

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

# Cartesian Plane

$$\mathbb{R}^2 = \{(x, y) | x \in \mathbb{R}, y \in \mathbb{R}\}$$



Can have  $\mathbb{R}^3, \mathbb{R}^4, \dots, \mathbb{R}^n$

## By the way

What is  $x \times x$ ?

What is  $x^2 \times x$ ?

What is  $x^2 \times x^2$ ?

More generally,  $x^n \times x^m = x^{m+n}$ .